

Mutation of Local Floer Chain Complexes across Hamiltonian Bifurcations

Dongho Lee

August 10th, 2026

Seoul National University, QSMS

QSMS Summer Workshop, Jeju.



Download slides

Contents

Based on a joint work with Hong-kwon Jo (SNU), arXiv:2607.22178.

Main result. We describe the explicit changes of the Floer chain complex for generic types of Hamiltonian bifurcations.

Based on a joint work with Hong-kwon Jo (SNU), arXiv:2607.22178.

Main result. We describe the explicit changes of the Floer chain complex for generic types of Hamiltonian bifurcations.

- **Motivation and Preliminaries.**

Bifurcations, the Conley–Zehnder index, and Floer homology.

- **Classification of Hamiltonian bifurcations.**

Birth-death, period doubling, phantom kiss and emission.

- **Chain complex mutation.**

Construction of cylinders, Morse–Bott cascade counting.

Motivation

Motivation 1. Bifurcations

A **bifurcation** is, heuristically, the birth or death of periodic orbits.

Several recent works [AFvK⁺23], [AB25], [JKvK26] relate bifurcations to symplectic invariants, for example Conley–Zehnder indices.

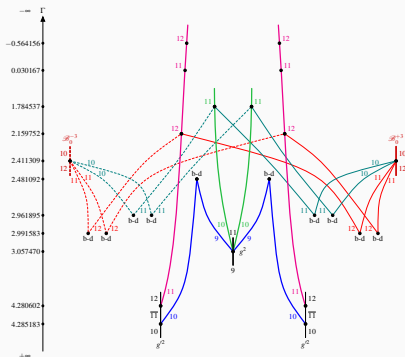


Figure 1: Bifurcation graph of Hill's lunar problem, [AB25].

Can we make a *symplectic dictionary* for bifurcation types?

Motivation 2. Rotating Kepler Problem

In the rotating Kepler problem, there are three important periodic orbits: retrograde, direct and vertical collision orbits.

The multiple covers of the retrograde and direct orbits bifurcate, producing orbits that look like flowers.

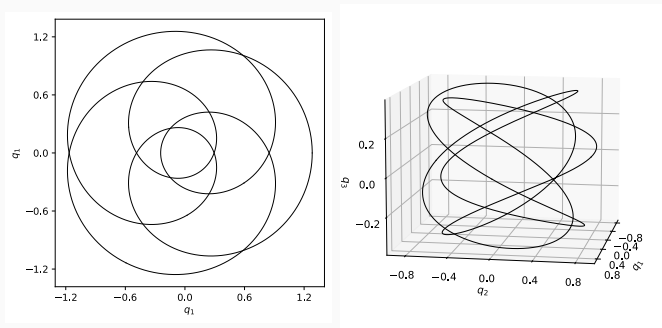
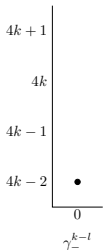
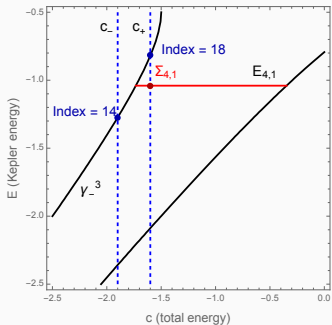


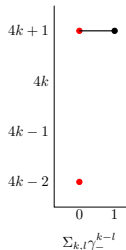
Figure 2: Resonant orbits of RKP, courtesy of Chankyu Joung.

Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed that this bifurcation can be translated into a change in the Floer chain complex.



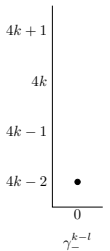
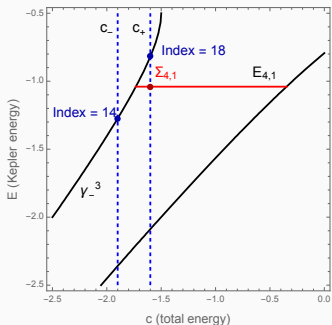
At $c_{k,l}^- - \epsilon$



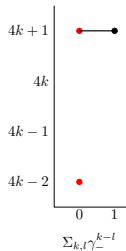
At $c_{k,l}^- + \epsilon$

Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed that this bifurcation can be translated into a change in the Floer chain complex.



At $c_{k,l}^- - \epsilon$



At $c_{k,l}^- + \epsilon$

Slogan. A bifurcation gives a *mutation of a Floer chain complex*.

We aim to describe this effect explicitly.

Periodic Orbits and Bifurcations

Periodic Orbits

Let (W, ω) be a symplectic manifold and $H : W \rightarrow \mathbb{R}$ be a function.

We define a **Hamiltonian vector field** by a relation $i_{X_H} \omega = -dH$.

Note. If $W = T^*\mathbb{R}^n$, $\omega = dp \wedge dq$ and $H = |p|^2/2 + V(q)$, X_H gives usual Hamiltonian equation in the classical mechanics,

$$\dot{q} = p, \quad \dot{p} = -V'(q).$$

Periodic Orbits

Let (W, ω) be a symplectic manifold and $H : W \rightarrow \mathbb{R}$ be a function.

We define a **Hamiltonian vector field** by a relation $i_{X_H} \omega = -dH$.

Note. If $W = T^*\mathbb{R}^n$, $\omega = dp \wedge dq$ and $H = |p|^2/2 + V(q)$, X_H gives usual Hamiltonian equation in the classical mechanics,

$$\dot{q} = p, \quad \dot{p} = -V'(q).$$

Main interest is **periodic orbits**, which is an orbit γ such that $\dot{\gamma} = X_H$ and $\gamma(0) = \gamma(T)$ for some $T > 0$.

Periodic orbits form backbone of the dynamics, and also connects the topology and dynamics. (Arnold conjecture, Floer theory)

Poincaré Section

Let γ be a periodic orbit of a Hamiltonian system and $p \in \gamma$.

A **Poincaré section** is a transversal section $\Sigma \subset H^{-1}(c)$ at p .

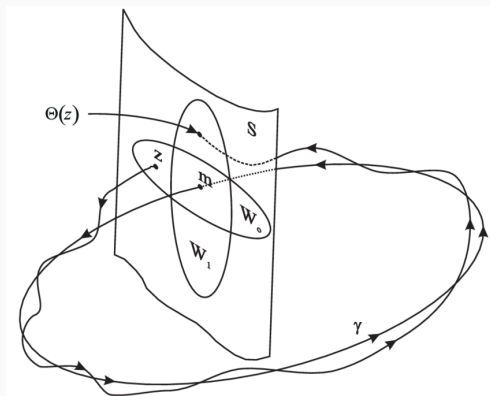


Figure 3: Poincaré section [OR99].

A **return map** is defined by $\Phi : U \subset \Sigma \rightarrow \Sigma$ by

$$\Psi(q) = Fl_{\tau(q)}^{X_H}(q), \quad \tau(q) = \min \left\{ t : Fl_t^{X_H}(q) \in \Sigma \right\}.$$

Floquet multipliers are eigenvalues of $d_p\Phi$.

Since $d_p\Phi$ is symplectic, these form a quadruple $\{\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}\}$.

A **return map** is defined by $\Phi : U \subset \Sigma \rightarrow \Sigma$ by

$$\Psi(q) = Fl_{\tau(q)}^{X_H}(q), \quad \tau(q) = \min \left\{ t : Fl_t^{X_H}(q) \in \Sigma \right\}.$$

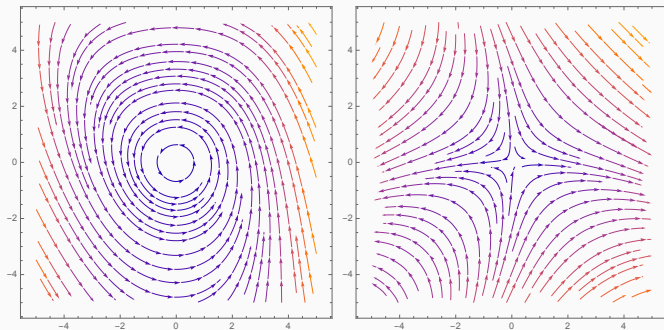
Floquet multipliers are eigenvalues of $d_p\Phi$.

Since $d_p\Phi$ is symplectic, these form a quadruple $\{\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}\}$.

In 4-dimensional autonomous case, we have 3 possibilities:

- $\lambda = \lambda^{-1} : \lambda \in S^1 \subset \mathbb{C}$, called **elliptic**.
- $\lambda = \bar{\lambda} : \lambda \in \mathbb{R} \setminus \{0\}$, called **hyperbolic**.
- $\lambda = \pm 1$.

We call this **stability** of a periodic orbit.



(a) Elliptic

(b) Hyperbolic

Figure 4: Elliptic and hyperbolic fixed points

Bifurcation

Let $H_\varepsilon : W^4 \rightarrow \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0 .

If $\lambda_0 \neq 1$, so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

This is a *continuation* of the orbit along the bifurcation parameter.

Bifurcation

Let $H_\varepsilon : W^4 \rightarrow \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0 .

If $\lambda_0 \neq 1$, so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

This is a *continuation* of the orbit along the bifurcation parameter.

We follow γ_ε and detect λ_ε . If $\lambda_\varepsilon = 1$ at some ε , then γ_ε is degenerate, and the orbit cylinder may break:

- periodic orbits may be born or vanish; this is called a **bifurcation**,
- if $\lambda_\varepsilon \neq 1$ but $\lambda_\varepsilon^k = 1$, then the k -th cover bifurcates.

Example. Period Doubling

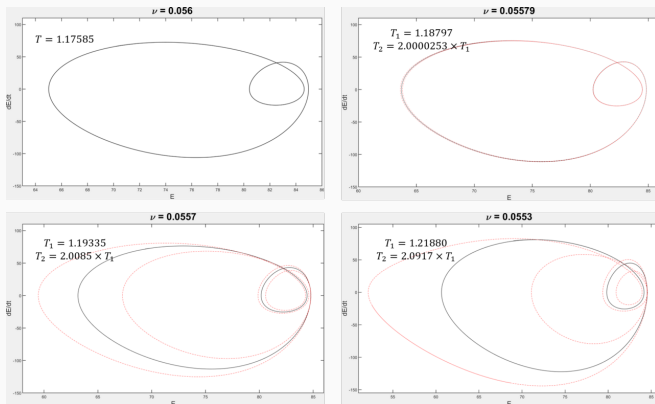


Figure 5: Period doubling in Kuramoto–Sivashinsky equation [Max21].

Period doubling is a generic bifurcation of a double cover.

The red dashed orbit is the new orbit that emerges from the bifurcation.

Conley–Zehnder Index

The **Conley–Zehnder index** $\mu_{\text{CZ}}(\gamma)$ is an integer associated with a periodic orbit and can be understood as a *symplectic winding number*.

We are interested in a change of index along the parameter; $\mu_{\text{CZ}}(\gamma_\epsilon)$.

Conley–Zehnder Index

The **Conley–Zehnder index** $\mu_{CZ}(\gamma)$ is an integer associated with a periodic orbit and can be understood as a *symplectic winding number*.

We are interested in a change of index along the parameter; $\mu_{CZ}(\gamma_\epsilon)$.

Along a non-degenerate family, the CZ index is constant.

Key idea. A change in μ_{CZ} = Degeneracy of γ_ϵ = A bifurcation occurs

Hence, a bifurcation can be detected via change of the CZ index.

Note. This is also useful in practical purposes. ([AB25], [JKvK26].)

Floer Homology

Floer homology $HF(W, H)$ is a topological invariant associated with a symplectic manifold W and Hamiltonian H . (cf. [AD14].)

This is determined by the **Floer chain complex** $CF(W, H)$, whose

- generators are 1-periodic orbits of H ,
- grading is given by CZ indices,
- the differential is determined by counting **Floer cylinder**, whose boundaries are asymptotic to periodic orbits.

Floer Homology

Floer homology $HF(W, H)$ is a topological invariant associated with a symplectic manifold W and Hamiltonian H . (cf. [AD14].)

This is determined by the **Floer chain complex** $CF(W, H)$, whose

- generators are 1-periodic orbits of H ,
- grading is given by CZ indices,
- the differential is determined by counting **Floer cylinder**, whose boundaries are asymptotic to periodic orbits.

Across a bifurcation, $CF(W, H)$ changes substantially;

- orbits are born or vanish $\Rightarrow$ the number of generators changes.
- the CZ index changes $\Rightarrow$ grading changes

However, the homology doesn't change, so the differential is responsible for keeping the homology invariant. *Can we describe this explicitly?*

Schematic Picture

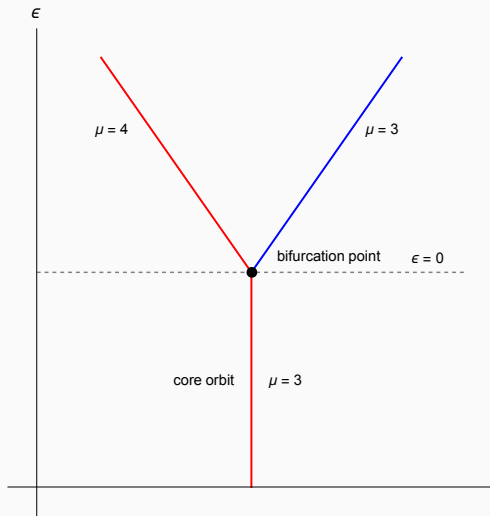


Figure 6: A bifurcation diagram for the period doubling.

Classification of Bifurcations

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number k of the underlying orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling, materialization (opposite stability types)
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number k of the underlying orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling, materialization (opposite stability types)
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Aside. We will use names from [AM78], where three extra types involving critical points are given.

Other names: touch-and-go, period tripling/quadrupling, etc.

Note. The fourth cover is a borderline case, depending on the coefficients of the normal form.

Birkhoff Normal Form

We find the fixed points of the iterated return map Φ_ε^k .

This can also be done in the essentially same way by finding critical points of the generating Hamiltonian G_ε of Φ_ε^k .

Birkhoff Normal Form

We find the fixed points of the iterated return map Φ_ε^k .

This can also be done in the essentially same way by finding critical points of the generating Hamiltonian G_ε of Φ_ε^k .

1. Using **Birkhoff normal form**, we can write the generating Hamiltonian of the return map Φ^k , i.e. $Fl_1^{X_G} = \Phi^k$, by

$$G_t = G_2 + \sum_{n \geq 3} G_{n,t}$$

where $\deg G_n = n$, $G_2^{ss} = \alpha(u^2 + v^2)$, $\dot{G}_{n,t} = \{G_2^{ss}, G_{n,t}\}$, $\lambda = e^{i\alpha}$.

2. If $\lambda^k = 1$, $G_2^{ss} = (2\pi l/k)(u^2 + v^2)$, which means that G is generated by polynomials commuting with rotation by $2\pi l/k$, i.e.,

$$G \in \mathbb{R}[\rho^2, \rho^k \cos(k\varphi - 2\pi lt), \rho^k \sin(k\varphi - 2\pi lt)].$$

3. Using rotating frame, we remove the time-dependency and G becomes an **autonomous $\mathbb{Z}_k$ -symmetric polynomials**.

Generating Hamiltonians – Birth-death

A generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 + \dots + \varepsilon(u + \dots).$$

- ($\varepsilon < 0$) There are no periodic orbits.
- ($\varepsilon > 0$) One elliptic and one hyperbolic orbit are born.

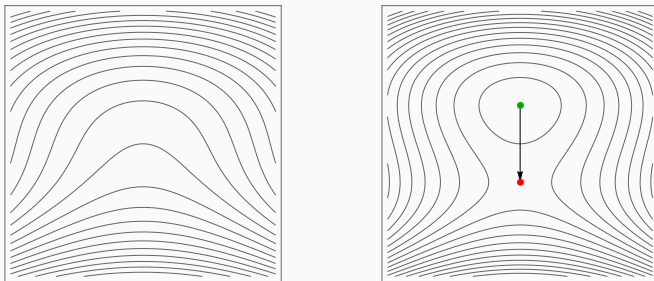


Figure 7: Critical points and gradient trajectories of birth-death bifurcation

Generating Hamiltonians – Period doubling

A generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 + \dots + \varepsilon(u^2 + \dots).$$

- ($\varepsilon < 0$) One elliptic orbit γ_0^2 exists.
- ($\varepsilon > 0$) γ_0^2 becomes hyperbolic, and a new elliptic orbit γ_E is born.

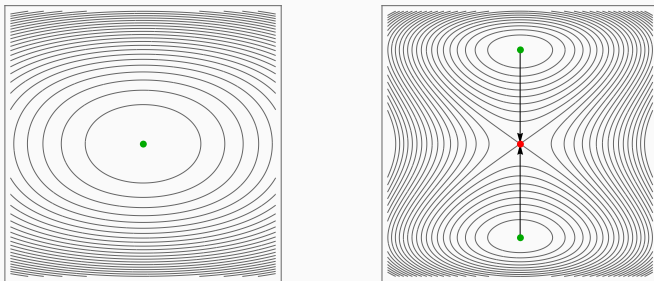


Figure 8: Period doubling bifurcation

Generating Hamiltonians – Phantom kiss

A generic 3rd or 4th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^k \cos k\theta + \dots + \varepsilon(r^2 + \dots).$$

- ($\varepsilon < 0$) One elliptic orbit γ_0^k and one hyperbolic orbit γ_H .
- ($\varepsilon = 0$) They meet, and suddenly become unstable.
- ($\varepsilon > 0$) The same configuration, but the index of γ_0^k changes.

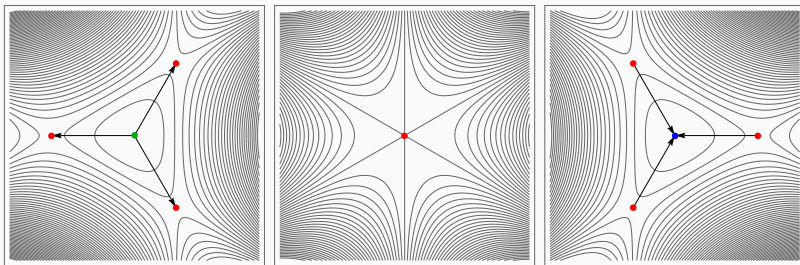


Figure 9: Phantom kiss bifurcation

Generating Hamiltonians – Emission

Generic $k \geq 4$ -th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^4 + \beta r^k \cos k\theta + \dots + \varepsilon(r^2 + \dots).$$

- ($\varepsilon < 0$) One elliptic orbit γ_0^k exists.
- ($\varepsilon > 0$) γ_0^k is still elliptic, and a new elliptic orbit γ_E and a new hyperbolic orbit γ_H are born.

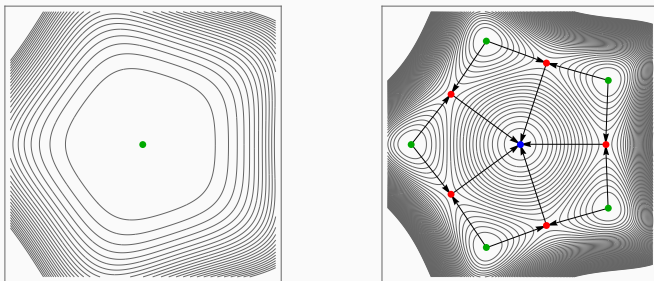


Figure 10: Emission bifurcation

Summary of the Classification

Name	Floquet Mult.	Before	Stability	Morse Index	After	Stability	Morse Index
Birth-death	$\lambda = 1$				γ_E	Elliptic	0 or 2
					γ_H	Hyperbolic	1
Period doubling	$\lambda = -1$	γ^2	Elliptic	0 or 2	γ^2	Hyperbolic	1
					γ_E	Elliptic	0 or 2
Materialization	$\lambda = -1$	γ^2	Hyperbolic	1	γ^2	Elliptic	0 or 2
					γ_H	Hyperbolic	1
Phantom kiss	$\lambda^k = 1$ $k = 3, 4$	γ^k	Elliptic	0 or 2	γ^k	Elliptic	2 or 0
		γ_H	Hyperbolic	1	γ_H	Hyperbolic	1
Emission	$\lambda^k = 1$ $k \geq 4$	γ^k	Elliptic	0 or 2	γ^k	Elliptic	2 or 0
					γ_H	Hyperbolic	1
					γ_E	Elliptic	0 or 2

Table 1: Generic Hamiltonian bifurcations

We also have $\text{Ind}(p) = 1 - \mu_{CZ}(\gamma)$, so *relative CZ index* is determined by the Morse index.

Note. We can distinguish the types further by the change of Morse index.

Bifurcation of Involutive Systems

Let a system have a symplectic $\mathbb{Z}_2$ -symmetry, and it inherits to the orbit and the return map.

The generating Hamiltonian of the return map of an orbit with $\lambda^k = 1$ now has an additional $\mathbb{Z}_2$ -symmetry, which becomes $\mathbb{Z}_{\text{lcm}(2,k)}$ -symmetric.

1. ($k = 1$) **Pitchfork**, which is the same as a period doubling.
2. ($k \geq 3$, odd) **Double emission**, which is the same as an emission.
3. (k even) Nothing changes.

Pitchfork

1. ($\varepsilon < 0$) One orbit γ_0 exists.
2. ($\varepsilon > 0$) γ_0 changes stability type, and two new orbits are born.

Double emission

1. ($\varepsilon < 0$) One elliptic orbit γ_0 exists.
2. ($\varepsilon > 0$) γ_0 is still elliptic. Two elliptic orbits $\gamma_{E,1,2}$ and two hyperbolic orbits $\gamma_{H,1,2}$ are born.

Chain Complex Mutation

An easy example. Birth-death bifurcation

Suppose an orbit γ_0 experiences the birth-death bifurcation along ε .

1. If $\varepsilon < 0$, there are no orbits.
2. If $\varepsilon > 0$, there are two orbits, say γ_H and γ_E .

An easy example. Birth-death bifurcation

Suppose an orbit γ_0 experiences the birth-death bifurcation along ε .

1. If $\varepsilon < 0$, there are no orbits.
2. If $\varepsilon > 0$, there are two orbits, say γ_H and γ_E .

Floer-theoretic viewpoint.

1. If $\varepsilon < 0$, $CF = 0$.
2. If $\varepsilon > 0$, $CF = \langle \gamma_H, \gamma_E \rangle$.
3. Both cases should produce the same (local) Floer homology.

An easy example. Birth-death bifurcation

Suppose an orbit γ_0 experiences the birth-death bifurcation along ε .

1. If $\varepsilon < 0$, there are no orbits.
2. If $\varepsilon > 0$, there are two orbits, say γ_H and γ_E .

Floer-theoretic viewpoint.

1. If $\varepsilon < 0$, $CF = 0$.
2. If $\varepsilon > 0$, $CF = \langle \gamma_H, \gamma_E \rangle$.
3. Both cases should produce the same (local) Floer homology.

Hence, $|\mu_{CZ}(\gamma_H) - \mu_{CZ}(\gamma_E)| = 1$ and they should cancel each other out, so there exists a Floer cylinder between them. (cf. [GG10].)

Note. Only **relative CZ index** is valid, since we're working on a neighborhood of a degenerate orbit $\cong S^1 \times \mathbb{R}^3$.

However, the situation is more complicated for higher covers.

An easy example. Birth-death bifurcation

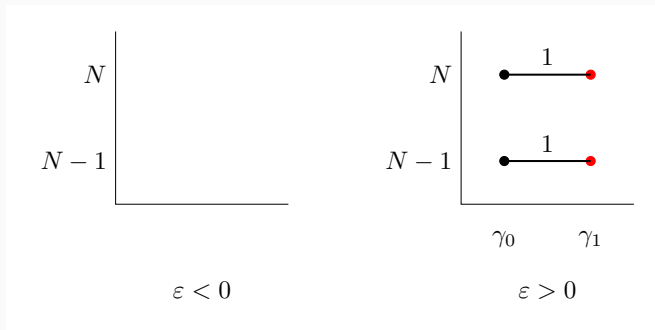


Figure 11: Chain complex mutation across the birth-death.

- Points at (n, m) corresponds to a generator of degree $n + m$.
- The line indicates the differential.
- If the number is odd, it means two generators cancel each other.
(We are using $\mathbb{Z}_2$ -coefficients.)

Main Result and Key Ideas

Main result. A full description of the mutation of Floer chain complex across the seven bifurcation types, and the local Floer homology of degenerate orbits.

Main Result and Key Ideas

Main result. A full description of the mutation of Floer chain complex across the seven bifurcation types, and the local Floer homology of degenerate orbits.

Key Ideas.

1. Construct a Hamiltonian H_ε represents a given bifurcation type.
2. Gradient trajectories of G_ε give Floer cylinders of H_ε .
3. Morse indices of critical points of G_ε give relative CZ indices.
4. We get cylinders connecting a simple orbit and a multiple cover.
 $\Rightarrow$ A contribution of 1 cylinder may not be exactly 1.
5. Using Morse–Bott cascades, we can compute the explicit chain complex and verify that its homology is invariant.

Model Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian and G_ε be a generating Hamiltonian for the k -th iterate of the return map.

We construct a model Hamiltonian on $I_r \times (\mathbb{R}/k\mathbb{Z})_\theta \times \Sigma_z$ by,

$$H_\varepsilon = kh(r) + G_\varepsilon(z).$$

where $h'(0) = 0$ and $h''(0) \neq 0$.

- H_ε is defined on the covering space of a neighborhood of a periodic orbit, but Floer-type information descends to the base space.
- The rotating number, or CZ index, is encoded in the covering map.
- H_ε is not exactly the same as $\tilde{H}_\varepsilon$, but has the same return map up to finite degree and Conley–Zehnder indices as $\tilde{H}_\varepsilon$.

Morse–Bott Cascade

Our periodic orbits have an S^1 -symmetry under time shifts, which makes the situation slightly different from the usual Floer theory.

We equip each orbit with a Morse function, and consider two critical points ($\max \hat{\gamma}$ and $\min \check{\gamma}$) as generators of a Floer chain complex.

In this case, the differential is given by counting **Morse–Bott cascades**; a sequence of Morse trajectories and Floer cylinders.

Morse–Bott Cascade

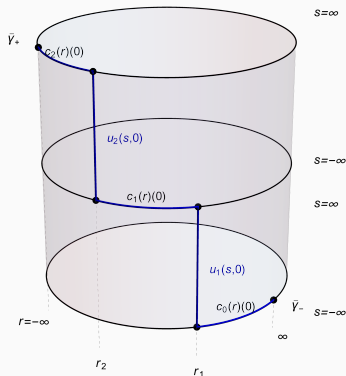
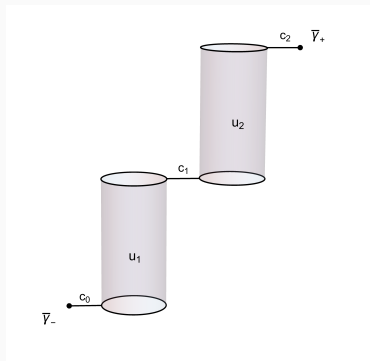


Figure 12: Morse–Bott cascades.

Cascade Counting Theorem

Theorem

Let $\gamma_{\pm}$ be two simple non-constant $1/k_{\pm}$ -periodic orbits, so that $\gamma_{\pm}^{k_{\pm}}$ are 1-periodic orbits. Moreover, assume that

$$\mu_{CZ}(\gamma_+^{k_+}) - \mu_{CZ}(\gamma_-^{k_-}) = 1.$$

Then, the quotient $\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1$ is a 0-dimensional $\mathbb{Z}_{(k_+, k_-)}$ -orbifold, and we have

$$\begin{aligned}\#_2 \mathcal{M}(\hat{\gamma}_-^{k_-}, \hat{\gamma}_+^{k_+}) &\equiv k_- \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod{2}, \\ \#_2 \mathcal{M}(\check{\gamma}_-^{k_-}, \check{\gamma}_+^{k_+}) &\equiv k_+ \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod{2}.\end{aligned}$$

In particular, if there is only one cylinder from a simple γ_0 to a multiple cover γ_1^k ,

$$\partial \hat{\gamma}_0 = k \hat{\gamma}_1^k, \quad \partial \check{\gamma}_0 = \check{\gamma}_1^k.$$

Idea of the proof

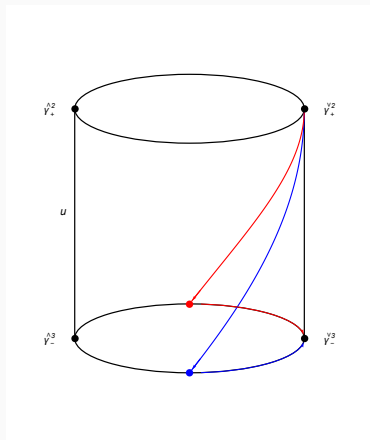
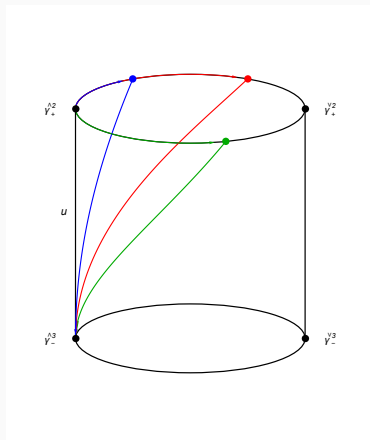


Figure 13: Cascades between 2nd and 3rd covers.

Chain complex mutation for Period doubling

1. If $\varepsilon < 0$, there is one orbit γ^2 .
2. If $\varepsilon > 0$, there are two orbits; γ^2 and γ_H .
3. $\mu_{CZ}(\gamma^2)$ changes, say, from N to $N - 1$.

Chain complex mutation for Period doubling

1. If $\varepsilon < 0$, there is one orbit γ^2 .
2. If $\varepsilon > 0$, there are two orbits; γ^2 and γ_H .
3. $\mu_{CZ}(\gamma^2)$ changes, say, from N to $N - 1$.

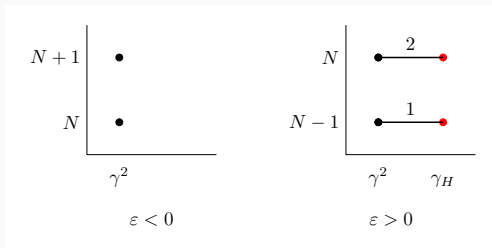


Figure 14: Chain complex mutation across the period doubling.

$\gamma_{H,\min}$ and $\gamma_{\min}^2$ cancel each other, while $\gamma_{H,\max}$ and $\gamma_{\max}^2$ do not.

$$HF_*^{\text{loc}}(\gamma_0^2) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^2, \\ 0 & \text{otherwise} \end{cases}$$

Chain complex mutations for Phantom kiss

1. If $\varepsilon < 0$, there are two orbits γ^k and γ_H^- .
2. If $\varepsilon > 0$, there are two orbits; γ^k and γ_H^+ .
3. $\mu_{CZ}(\gamma^k)$ changes, say, from $N + 1$ to $N - 1$.
4. $\mu_{CZ}(\gamma_H)$ doesn't change.

Chain complex mutations for Phantom kiss

1. If $\varepsilon < 0$, there are two orbits γ^k and γ_H^- .
2. If $\varepsilon > 0$, there are two orbits; γ^k and γ_H^+ .
3. $\mu_{CZ}(\gamma^k)$ changes, say, from $N + 1$ to $N - 1$.
4. $\mu_{CZ}(\gamma_H)$ doesn't change.

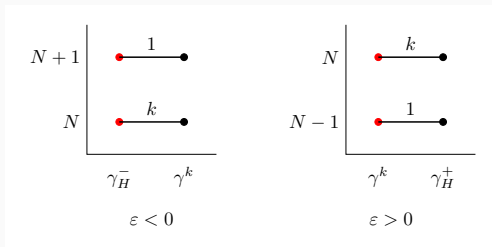


Figure 15: Chain complex mutations across the phantom kiss

$$HF_*^{\text{loc}}(\gamma_0^3) = 0, \quad HF_*^{\text{loc}}(\gamma_0^4) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = -1, 0 \text{ relative to } \gamma_-^4, \\ 0 & \text{otherwise} \end{cases}$$

Chain complex mutations for Emission

1. If $\varepsilon < 0$, there is one orbit γ^k .
2. If $\varepsilon > 0$, there are three orbits; γ^k , γ_H , γ_E .
3. $\mu_{CZ}(\gamma^k)$ changes, say, from $N + 1$ to $N - 1$.

Chain complex mutations for Emission

1. If $\varepsilon < 0$, there is one orbit γ^k .
2. If $\varepsilon > 0$, there are three orbits; γ^k , γ_H , γ_E .
3. $\mu_{CZ}(\gamma^k)$ changes, say, from $N + 1$ to $N - 1$.

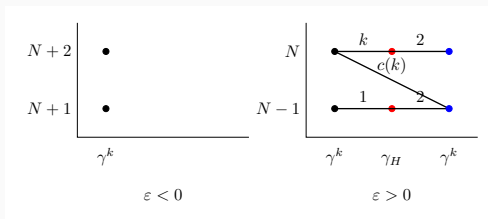


Figure 16: Chain complex mutations across the emission

$c(k)$ depends on the choice of Morse function, but always odd if k is even.

$$HF_*^{\text{loc}}(\gamma_0^k) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^k, \\ 0 & \text{otherwise} \end{cases}$$

Chain complex mutations for Emission

To find $c(k)$, we need to count the cascade of length 2 from $\tilde{\gamma}^k$ to $\hat{\gamma}_E$, which is equivalent to a count of Morse trajectories

Image of $\tilde{\gamma}^k$ on $\gamma_H \rightarrow$ Preimage of $\hat{\gamma}_E$ on γ_H .

This falls into three cases, depending on the choice of the Morse function.

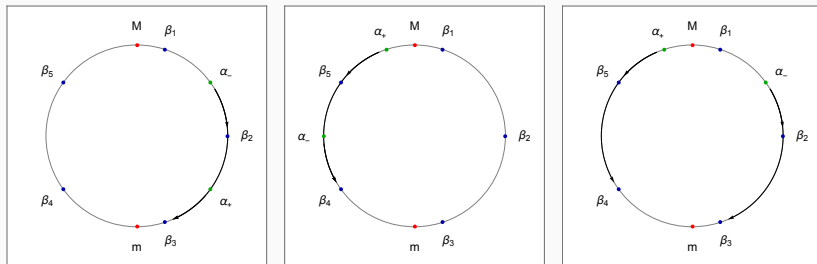


Figure 17: Three cases of $c(k)$.

Chain complex mutations for Pitchfork

1. If $\varepsilon < 0$, there is one orbit γ .
2. If $\varepsilon > 0$, there are three orbits; $\gamma, \gamma_1, \gamma_2$.
3. $\mu_{CZ}(\gamma^k)$ changes, say, from N to $N - 1$.

Chain complex mutations for Pitchfork

1. If $\varepsilon < 0$, there is one orbit γ .
2. If $\varepsilon > 0$, there are three orbits; γ , γ_1 , γ_2 .
3. $\mu_{CZ}(\gamma^k)$ changes, say, from N to $N - 1$.

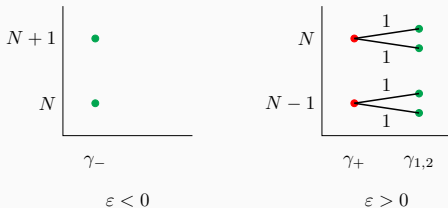


Figure 18: Chain complex mutations across the pitchfork

$$HF_*^{\text{loc}}(\gamma_0) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-, \\ 0 & \text{otherwise} \end{cases}$$

Chain complex mutations for Emission

1. If $\varepsilon < 0$, there is one orbit γ^k .
2. If $\varepsilon > 0$, there are five orbits; γ^k , $\gamma_{H,1,2}$, $\gamma_{E,1,2}$.
3. $\mu_{CZ}(\gamma^k)$ changes, say, from $N + 1$ to $N - 1$.

Chain complex mutations for Emission

1. If $\varepsilon < 0$, there is one orbit γ^k .
2. If $\varepsilon > 0$, there are five orbits; γ^k , $\gamma_{H,1,2}$, $\gamma_{E,1,2}$.
3. $\mu_{CZ}(\gamma^k)$ changes, say, from $N + 1$ to $N - 1$.

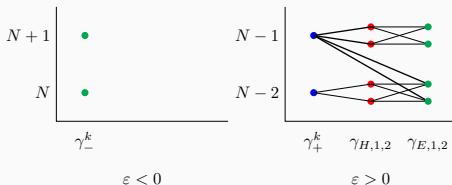


Figure 19: Chain complex mutations across the emission

$$HF_*^{\text{loc}}(\gamma_0^k) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^k, \\ 0 & \text{otherwise} \end{cases}$$

Further discussions

- Orientation and $\mathbb{Z}$ -coefficients.
- More symmetries, multi-parameter, higher dimensional bifurcations.
- This gives a *generic* change of local Floer chain complex.
⇒ Persistent module description and Floer-type Cerf theory?
- Bifurcation of symmetric orbits can be described as a Lagrangian Floer theory ⇒ Similar work for Lagrangians?



Cengiz Aydin and Alexander Batkhin.

Studying network of symmetric periodic orbit families of the Hill problem via symplectic invariants.

Celestial Mech. Dynam. Astronom., 137(2):Paper No. 12, 77, 2025.



Michèle Audin and Mihai Damian.

Morse theory and Floer homology.

Universitext. Springer, London; EDP Sciences, Les Ulis, 2014.

Translated from the 2010 French original by Reinie Ern .



Cengiz Aydin, Urs Frauenfelder, Otto van Koert, Dayung Koh, and Agustin Moreno.

Symplectic geometry and space mission design.

2023.

References ii



Ralph Abraham and Jerrold E. Marsden.

Foundations of mechanics.

Benjamin/Cummings Publishing Co., Inc., Advanced Book Program,
Reading, MA, second edition, 1978.



Viktor L. Ginzburg and Basak Z. Gurel.

Local floer homology and the action gap.

J. Symplectic Geom., 8(3):323–357, 2010.



Chankyu Joung, Dayung Koh, and Otto van Koert.

**Bifurcations of highly inclined near halo orbits using moser
regularization, 2026.**



Dongho Lee.

Conley–zehnder indices of spatial rotating kepler problem.

Journal of Topology and Analysis, 0(0):1–40, 2026.



MaxwellMolecule.

K-s equation period doubling.

Wikimedia Commons, February 2021.

Illustration of period doubling in the Kuramoto–Sivashinsky equation. CC0 1.0 Universal Public Domain Dedication.



K. R. Meyer.

Generic bifurcation of periodic points.

Transactions of the American Mathematical Society, 149(1):95–107, 1970.



Juan-Pablo Ortega and Tudor Ratiu.

A dirichlet criterion for the stability of periodic and relative periodic orbits in hamiltonian systems.

Journal of Geometry and Physics, 32:131–159, 12 1999.



Figure 20: Thank you for your attention!

Questions, comments, and further discussion are welcome!